

NIDS Session Report

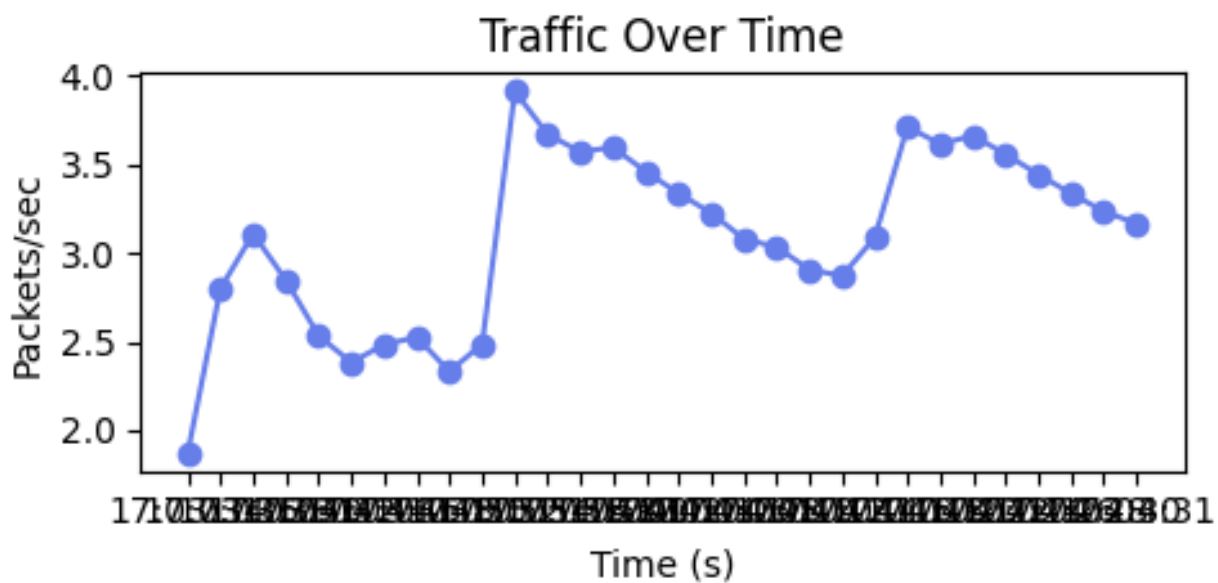
AI-Based Network Intrusion Detection System

This report summarizes your network activity and any suspicious events detected. It's designed to be easy to understand, even if you're not a network expert.

How to Read This Report

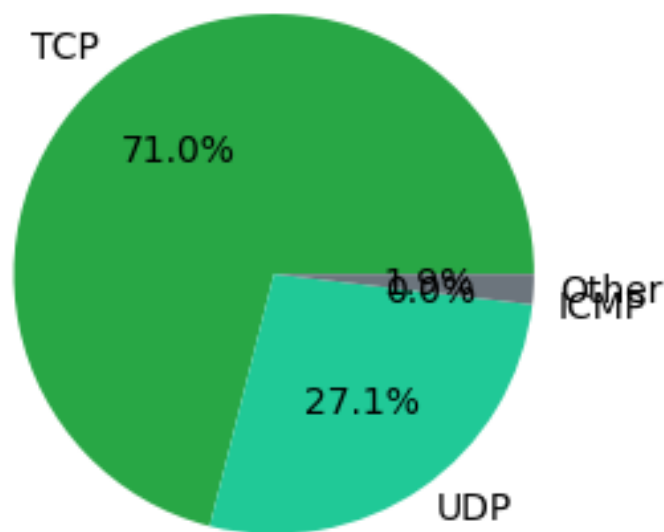
- Packets: Each piece of data sent over your network.
- Anomaly: Something unusual or unexpected in the network traffic.
- Alert: A warning about a specific suspicious activity.

Traffic Over Time



Protocol Distribution

Protocol Distribution



Session Summary

Start Time:	1750246404.3075306
Stop Time:	2025-06-18T17:04:31.874014
Uptime:	67 seconds
Packets Processed:	214
Anomalies Detected:	13
Packet Rate:	3.17 packets/sec

Protocol Statistics

TCP PACKETS:	152
UDP PACKETS:	58
ICMP PACKETS:	0
OTHER PACKETS:	4

Security Alerts (4)

Type: MI Anomaly | Time: 2025-06-18T17:03:29.349301

Details: {'anomaly_score': -7.309258797877921, 'features': {'packet_count': 13, 'total_bytes': 780, 'avg_packet_length': np.float64(60.0), 'std_packet_length': np.float64(0.0), 'min_packet_length': 60, 'max_packet_length': 60, 'avg_payload_size': np.float64(2.0), 'std_payload_size': np.float64(0.0), 'total_payload_size': 26, 'tcp_count': 13, 'udp_count': 0, 'icmp_count': 0, 'other_protocol_count': 0, 'protocol_diversity': 1, 'tcp_ratio': 1.0, 'udp_ratio': 0.0,

'icmp_ratio': 0.0, 'other_protocol_ratio': 0.0, 'unique_src_ports': 1, 'unique_dst_ports': 13, 'total_ports_accessed': 14, 'common_port_ratio': 0.0, 'high_port_ratio': 1.0, 'avg_dst_port': np.float64(54030.38461538462), 'std_dst_port': np.float64(6.208411930455713), 'min_dst_port': 54023, 'max_dst_port': 54047, 'unique_connections': 7, 'total_connection_attempts': 13, 'avg_packets_per_connection': 1.8571428571428572, 'connection_diversity': 0.5384615384615384, 'max_packets_per_connection': 2, 'unique_src_ips': 7, 'unique_dst_ips': 1, 'src_ip_diversity': 0.5384615384615384, 'dst_ip_diversity': 0.07692307692307693, 'private_src_ip_ratio': 0.0, 'private_dst_ip_ratio': 1.0, 'packet_rate': 2.8924822767270775, 'inter_arrival_time_mean': np.float64(0.37453413009643555), 'inter_arrival_time_std': np.float64(0.5439098990347546), 'burst_count': 6, 'burst_ratio': 0.5, 'payload_size_mean': np.float64(2.0), 'payload_size_std': np.float64(0.0), 'zero_payload_ratio': 0.0, 'large_payload_ratio': 0.0, 'payload_size_entropy': 0, 'syn_count': 13, 'fin_count': 0, 'rst_count': 0, 'psh_count': 0, 'ack_count': 13, 'urg_count': 0, 'syn_ratio': 1.0, 'fin_ratio': 0.0, 'rst_ratio': 0.0, 'psh_ratio': 0.0, 'ack_ratio': 1.0, 'urg_ratio': 0.0}, 'timestamp': '2025-06-18T17:03:29.349286', 'packet_count': 13, 'protocol_distribution': {'tcp': 13, 'udp': 0, 'icmp': 0}, 'connection_info': {'unique_connections': 7, 'unique_src_ips': 7, 'unique_dst_ips': 1}}

Explanation: The AI detected network activity that looks very different from normal. This could mean someone is trying something unusual or potentially harmful on your network.

Type: Port Scan | Time: 2025-06-18T17:03:30.862986

Details: {'src_ip': '10.0.2.15', 'dst_ip': '3.229.150.19', 'dst_port': 443, 'protocol': 6}

Explanation: A device tried to quickly check many different ports on another device. Hackers often do this to find weak spots before launching an attack.

Type: Ddos Indicator | Time: 2025-06-18T17:04:17.121865

Details: {'src_ip': '35.168.50.176', 'dst_ip': '10.0.2.15', 'packet_size': 1494, 'protocol': 6}

Explanation: A large number of packets were sent to a single device, possibly overwhelming it. This is a common sign of a Distributed Denial of Service (DDoS) attack, which can take down services.

Type: MI Anomaly | Time: 2025-06-18T17:04:29.575544

Details: {'anomaly_score': -7.30022824337124, 'features': {'packet_count': 100, 'total_bytes': 16093, 'avg_packet_length': np.float64(160.93), 'std_packet_length': np.float64(311.499157462745), 'min_packet_length': 42, 'max_packet_length': 1536, 'avg_payload_size': np.float64(106.18), 'std_payload_size': np.float64(312.64495454108965), 'total_payload_size': 10618, 'tcp_count': 82, 'udp_count': 14, 'icmp_count': 0, 'other_protocol_count': 0, 'protocol_diversity': 2, 'tcp_ratio': 0.8541666666666666, 'udp_ratio': 0.14583333333333334, 'icmp_ratio': 0.0, 'other_protocol_ratio': 0.0, 'unique_src_ports': 19, 'unique_dst_ports': 27, 'total_ports_accessed': 27, 'common_port_ratio': 0.3333333333333333, 'high_port_ratio': 0.6666666666666666, 'avg_dst_port': np.float64(36215.291666666664), 'std_dst_port': np.float64(25390.995129473176), 'min_dst_port': 53, 'max_dst_port': 62471, 'unique_connections': 30, 'total_connection_attempts': 96, 'avg_packets_per_connection': 3.2, 'connection_diversity': 0.3, 'max_packets_per_connection': 14, 'unique_src_ips': 18, 'unique_dst_ips': 14, 'src_ip_diversity': 0.18, 'dst_ip_diversity': 0.14, 'private_src_ip_ratio': 0.11111111111111111, 'private_dst_ip_ratio': 0.14285714285714285, 'packet_rate': 3.3306544181371023, 'inter_arrival_time_mean': np.float64(0.3032740366579306), 'inter_arrival_time_std': np.float64(0.7799106488348029), 'burst_count': 51,

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'burst_ratio':      0.5151515151515151,      'payload_size_mean':      np.float64(106.18),      'payload_size_std':  
np.float64(312.64495454108965), 'zero_payload_ratio':  0.19, 'large_payload_ratio':  0.05, 'payload_size_entropy':  
np.float64(0.8309464733646963), 'syn_count': 35, 'fin_count': 2, 'rst_count': 8, 'psh_count': 15, 'ack_count': 74,  
'urg_count': 0, 'syn_ratio': 0.4268292682926829, 'fin_ratio': 0.024390243902439025, 'rst_ratio': 0.0975609756097561,  
'psh_ratio':      0.18292682926829268,      'ack_ratio':      0.9024390243902439,      'urg_ratio':      0.0},      'timestamp':  
'2025-06-18T17:04:29.575531', 'packet_count': 100, 'protocol_distribution': {'tcp': 82, 'udp': 14, 'icmp': 0},  
'connection_info': {'unique_connections': 30, 'unique_src_ips': 18, 'unique_dst_ips': 14}}
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Explanation: The AI detected network activity that looks very different from normal. This could mean someone is trying something unusual or potentially harmful on your network.

What Should I Do?

- If you see many alerts: Consider checking your devices for unusual activity or malware.
- If you see DDoS or Port Scan alerts: Make sure your firewall is active and up-to-date.
- If unsure: Share this report with your IT support or a cybersecurity professional.

Glossary of Terms

Packets Processed:	Total number of network packets analyzed by the NIDS during this session.
Anomalies Detected:	Number of network events flagged as suspicious or abnormal by the NIDS.
Packet Rate:	Current rate at which packets are being processed.
Uptime:	How long the NIDS was running.
TCP/UDP/ICMP/Other:	Number of packets for each protocol.
Port Scan:	A port scan was detected. This means a host is rapidly probing multiple ports on a target, which is often a precursor to an attack.
DDoS Indicator:	Possible Distributed Denial of Service (DDoS) activity detected. This means a large number of packets are being sent to a target, which may overwhelm the system.
Unusual Protocol:	Unusual network protocol detected. This could indicate suspicious or unauthorized network activity.
ML Anomaly:	The machine learning model detected traffic that is statistically unusual compared to normal patterns.